

Vishesh Raju

201-936-4203 | vr432@njit.edu | linkedin.com/in/vishesh-raju | github.com/Vishesh0-7

EDUCATION

New Jersey Institute of Technology

Masters in Computer Science

Newark, New Jersey

Aug 2024 – May 2026

- Current GPA: **3.9/4.0**
- Relevant Coursework: Machine Learning, Database Management and System Design, Big Data, Data Structures and Algorithms, Data Mining, Data Analytics, Deep Learning, Operating Systems, Web Development

R N Shetty Institute of Technology

Bachelor of Technology in Artificial Intelligence and Machine Learning

Bengaluru, Karnataka

Dec 2020 – May 2024

- Cumulative GPA: **7.44/10.0**
- Relevant Coursework: Algorithms and Data Structure, Database Management System, Operating System, Machine Learning, Artificial Intelligence, Neural Networks and Deep Learning, Digital Image Processing

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL, R
- **Technologies/Frameworks:** HTML5, CSS3, Hadoop, AWS, Apache Oozie, PowerBI, PyTorch, Tensorflow
- **Developer Tools:** VS Code, PyCharm, Eclipse, Jupyter, RStudio, ImageJ

WORK EXPERIENCE

Research Assistant

New Jersey Institute of Technology - CIBM3

May 2025 – Present

Newark, New Jersey

- Analyzing rodent brain tissue images to identify and quantify microglial morphologies across species.
- Working with interdisciplinary teams on AI-driven methods for neuroinflammation pattern analysis.

Student Intern

22by7 Technologies Ltd.

Oct 2023 – Dec 2023

Bengaluru, Karnataka

- Facilitated development of a predictive maintenance project during a **2-month** internship.
- Demonstrated skills in data analysis and project management, gaining practical insights into tech applications.
- Executed project tasks contributing to a model with **97% accuracy**.

PROJECTS

Chicago Crime Analysis

| *BIG DATA*

Technologies: Hadoop, Oozie, MapReduce, Python, AWS

- Processed over **8 million records** from the Chicago Crime Dataset using Hadoop and Oozie workflows.
- Developed **3 optimized MapReduce jobs** for crime trend and hotspot analysis.
- Managed up to **12 scalable AWS Virtual Machines** for efficient big data processing.

Text to Face Generation

| *CAPSTONE*

Technologies: GANs, ControlGAN, Python, PyTorch, TextCUFSF Dataset

- **Led a team of 4** to develop a text-to-face generation system using GANs for realistic facial image synthesis.
- Implemented ControlGAN on the TextCUFSF dataset, which includes **1,139 photo-sketch pairs**.
- Achieved an **FID score of 32.02** on the test dataset, validating model performance.

Attendance System with Face Recognition

| *COMPUTER VISION*

Technologies: OpenCV, Dlib, Face Recognition, Flask, NumPy, Pandas

- Built a face recognition-based attendance system using OpenCV, Dlib, and Face Recognition libraries.
- Utilized NumPy and Pandas for efficient data handling and analysis of attendance logs.
- Deployed the system using Flask to enable real-time tracking and usability.